Assignment 3

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Student Name/Number: (doubleclick to edit)

Instructions

- Edit the above cell to include your name and student number.
- Submit just this `ipynb` to Canvas. Do not rename, it associates your student number with the submission automatically.

```
using Distributions, Plots, LaTeXStrings, LinearAlgebra, Statistics, Random
```

Question 1

Following the notes on AR(1) processes rather than plotting the distribution as normal instead lets see what the stationary distribution looks like with simulation.

Part (a)

From $X_0 = 1.0$ simulate up to $T = 1000$ using the process

$$X_t = aX_{t-1} + b + cW_t$$

Where $a = 0.9, b = 0.1, c = 0.5$.

```
T = 1000
X_0 = 1.0
a = 0.9
b = 0.1
c = 0.5
# Add code here
```

```
0.5
```

Part (b)

On the same graph plot the histogram of those simulated values (i.e., $\{X_0, \dots X_T\}$) then plot the density of the stationary distribution calculated in closed from in those notes (i.e. create a normal distribution with $\mu^* = b/(1-a)$ and $v^* = c^2/(1-a^2)$)

Hint: `histogram(X, normed=true)` normalizes the empirical draws so they are a proper PMF.

```
# Add code here
```

Part (c)

What happens if you discard the first 200 observations from that simulation (i.e., $\{X_{199}, \dots X_T\})$)?

```
# Add code here
```

Do these line up approximately? Explain why it may be better or worse?

(double click to edit your answer)

Reusable functions, do not modify without clearly annotating your changes.

```
function lorenz(v)  # assumed sorted vector
    S = cumsum(v)  # cumulative sums: [v[1], v[1] + v[2], ... ]
    F = (1:length(v)) / length(v)
    L = S ./ S[end]
    return (; F, L) # returns named tuple
end
# Assumes that v is sorted!
gini(v) = (2 * sum(i * y for (i,y) in enumerate(v))/sum(v)
           - (length(v) + 1))/length(v)
```

```
gini (generic function with 1 method)
```

Question 2

You can create a Pareto distribution with tail parameter `a` and draw from it with

```
a = 1.5
d = Pareto(a)
rand(d, 3)
```

```
3-element Vector{Float64}:
 4.142254953345905
 1.1598547705484767
 1.1752588139237106
```

The gini coefficient of a Pareto distribution is given by

$$G = \frac{1}{2a-1}$$

Part (a)

Draw `N=1000` observations from the Pareto distribution for `a = 1.5` and calculate the gini coefficient. Compare to the closed-form solution

```
N = 1000
a = 1.5
# your code here
```

```
1.5
```

Part (b)

Given a pdf $f(x)$ the cumulative distribution function is given by $F(x) = \int_{-\infty}^{x} f(y)dy$. The complementary cumulative distribution function (CCDF) is given by $F^c(x) = 1 - F(x)$. It starts at 1 at the minimum of the support of the distribution and then drops to 0 at the maximum of the support.

Take the following code which calculates the CCDF for a distribution and a range of values `x`.

For this, plot the relationship between the `log(x)` and `log(ccdf(d,x))` for the distribution below (this is called a log-log plot). Describe what you know about this relationship?

```
ccdf(d, x) = 1 - cdf(d, x)

N = 1000
a = 1.5
d = Pareto(a)
max_x = 10.0
num_points = 100
x = range(support(d).lb, max_x, num_points)  # from the lower bound of support to max_x

# edit your code here
```

```
1.0:0.09090909090909091:10.0
```

Part (b)

For the above case, numerically calculate the slope of this line (easiest is just rise over run). Compare it to the tail parameter `a`.

```
# your code here
```

Part (c)

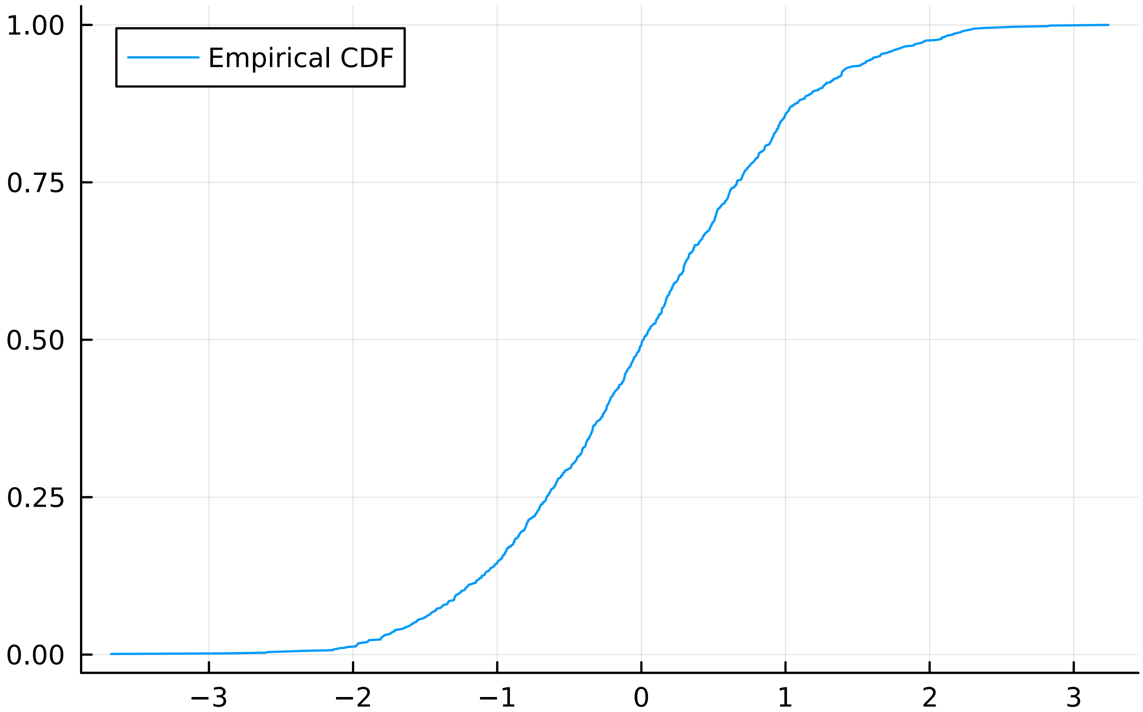
Do we will do the empirical counterpart. Given unweighted vector $\{X_n\}_{n=1}^N$ of observations, we can define the empirical CDF as

$$\hat{F}(x) = \frac{\text{number of observations } X_n \leq x}{N}$$

With the equivalent CCDF as $1 - \hat{F}(x)$.

The code below calculates the empirical CDF for some simulated data. Adapt it to (1) draw `N` draws from the Pareto distribution above with `a = 1.5`; (2) plot the same log-log plot as above; and (3) overlay the plot with the theoretical log-log plot from the previous part to compare the two.

```
# your code to adapt
N = 1000
x = sort(randn(N))  # this draws normals, adapt to our areto
F_hat(x) = (1:length(x)) ./ length(x) # the cdf values at the x values are just the counts no
plot(x, F_hat(x), label="Empirical CDF") # adapt to do the log-log plot
```



This shows methods for understanding the tail behavior of distributions. If it approaches a straight line, then it is called a power-law tail.

Part (d)

Now lets do the same empirical check on the tail behavior with a LogNormal distribution - which does not have a power-law tail and the Frechet distribution - which does.

First, adapt your code above to (1) draw `N=1000` elements from the `LogNormal(0.0, 0.5)` (2) plot the pdf to get a feel for the distribution on a reasonable range, then finally (3) plot the theoretical and empirical log-log plots as you did before.

```
# your code to adapt
N = 1000
d = LogNormal(0.0, 0.5)
# add code here to plot the pdf
# add code for the log-log plot
```

```
LogNormal{Float64}( =0.0,  =0.5)
```

```
# your code to adapt
# add code for the log-log plot
```

Next do the same thing except with a `Frechet(1.5, 1)` (which has a tail parameter of 1.5).

```
# your code to adapt
N = 1000
d = Frechet(1.5, 1.0)
# add code here to plot the pdf
# add code for the log-log plot
```

```
Frechet{Float64}( =1.5,  =1.0)
```

```
# your code to adapt
# add code for the log-log plot
```

Part (e)

Comparing your answers in Part (d) and (c) what do you notice is different theoretically between the log-log plots of distributions with and without power-law tails? Play around with it and see if you feel you could just look at an empirical plot and tell if it is a power-law tail or not?

(double click to edit your answer)